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Sensing Data

Towards ethnographic methods for data positionality

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Sensing data: Towards ethnographic methods for data positionality

Mette Simonsen Abildgaard

Abstract

How can we develop an embodied sense of data in a world in which digital connectivity is distributed unevenly, yet increasingly taken for granted? This article introduces 'sensing data' as an ethnographic method for engaging with data not as abstract flows, but as material, situated and sensory phenomena. Extending the study of 'mundane data' towards infrastructure studies, this method invites careful attention to how kilobytes, megabytes and gigabytes circulate in everyday entanglements with mobile phones, undersea cables and satellite connections. Sensing data is presented in two iterations: First, an autoethnographic account of fieldwork in North Greenland's so-called 'satellite zone' foregrounds data as objects of care, discipline and contestation. Second, a teaching exercise involving severe data restrictions draws attention to the infrastructural systems shaping everyday digital practices. Together, these experiments propose sensing data as a way to articulate 'data positionality'– an embodied and situated awareness of the conditions under which data are made, moved and experienced. While 'unlimited' data plans render infrastructure seemingly frictionless, sensing data insists on the importance of cultivating felt knowledge of data as planetary and political.

Keywords

Sensing data, situated data, mundane data, ethnographic methods, infrastructure, telecommunication

Introduction

> Day 9, 17/8: 14 GB of 20 used (9.1 on hot spot, 1.7 on app store)
>
> *Am getting stressed out about my data consumption. I spent 14 GB in 7 days and will be here 8 more days. My problem after having downloaded Civilization is when my computer is using my phone's hot spot. I spent 1.6 GB on that. Probably when all the mail and messages are downloaded again. And things like 'system services' in my phone use over 0.5 GB. What even is that? They mention 'security services??'*

In this article, I introduce 'sensing data' as an ethnographic methodology for engaging with data as infrastructural and situated phenomena. The aim is not only to explore how researchers can attune to data's material presence in systems such as mobile networks and satellite connections, but also to propose sensing data as a method for teaching and learning about digital infrastructures. I do so by drawing three interconnected lines together: an autoethnographic account of limited digital capacity during fieldwork in Greenland's satellite zone; a pedagogical experiment that implements the method in a teaching context; and development of sensing data as a methodological concept grounded in phenomenological infrastructure studies. Taken together, these parts demonstrate how 'data positionality' – that is our embodied and emplaced (Pink, 2011) sense of data's character and shape – can be cultivated through reflexive, sensorial engagements with data in everyday life.

In ongoing discourse about so-called computational or machine anthropology and the complementarity between 'thick' qualitative and 'big' quantitative data, several authors rightly have suggested that these approaches are not mutually exclusive, but can support and enrich each other (Albris et al., 2021; Munk et al., 2022; Pedersen, 2023). However, the big and small data I focus on here are not computational techniques meshing with firsthand ethnographic observation, or even data sharing practices as such. Instead, I focus on data in information systems in the form of kilobytes and zettabytes circulating among

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smartphones, tablets, laptops, Internet routers and larger telecommunication infrastructures, such as fibre-optic cables and satellites, that enable this circulation, and our individual relationship, with this data.

Concretely, I want to talk about how to engage with digital information systems data as situated phenomena in ethnographic work. Of course, ethnographic research already is preoccupied with data. For example, ethnographers use the term *ethnographic data* in reference to information that has been collected during ethnographic observations or interviews. In this way, all fieldwork produces data, and *data* is used habitually as a synonym for empirical material, i.e., knowledge that stems from the ethnographer's own experience and senses. Furthermore, when we talk about data from empirical studies, we often contrast it with 'a priori knowledge' or 'data of reason' that arrive not from direct observations, but from what we know innately or from logical inference (Bealer, 2000). So, *data* already is an often-used term in ethnography, with ethnographic data being a central, but contradictory, concept that on one hand points to knowledge we have from direct observation, while on the other includes the possibility that it stems from assumptions or logical deductions. In this paper, I aim to discuss how we can connect these ethnographic practices of producing and engaging with data, with practices of relating to data as bytes flowing in telecommunication systems. While bytes are measured in highly standardised, globally uniform units, they also travel in infrastructural systems that are vastly different worldwide. So, how can we study this kind of data empirically and phenomenologically?

The introductory excerpt is from my field notes taken in 2022 during fieldwork in Qaanaaq, Greenland's northernmost town. Qaanaaq's digital access to the rest of the world runs through a 175 Mbit/s geostationary satellite connection that the town shares with nearby Pituffik Space Base (formerly Thule Air Base); surrounding settlements Savissivik, Siorapaluk and Qeqertat; and Nerlerit Inaat Airport (Abildgaard, 2024). As a researcher with longtime ethnographic engagement with Greenland's telecommunication infrastructures, I was in Qaanaaq to follow ongoing work to upgrade the town's ground satellite station before shifting to HISPASAT, a newer geostationary satellite. Once linked to Qaanaaq's satellite dish, HISPASAT promised Qaanaaq's inhabitants a faster and more stable Internet connection, and for the first time, the opportunity to acquire a flat rate home Internet subscription through a 4G modem (Abildgaard, 2024). The absence, then as well as now, of flat-rate 'unlimited data' mobile subscriptions in Qaanaaq – encompassing talk, text and data – was unfamiliar to me, a lifelong resident of Copenhagen, Denmark's capital city in Northern Europe. My everyday routines usually include high-speed Wi-Fi Internet at work, an abundance of free hot spots around town and an inexpensive phone plan with unlimited talk, texting and data. To get a better sense of what Qaanaaq's telecommunication infrastructure means for life in this town, an autoethnographic exercise with data became an integral part of my fieldwork in Qaanaaq. The exercise involved a daily log of my data use, reflections on these uses and my diverse strategies managing data in an unfamiliar situation with limited capacity.

In this article, I introduce and discuss the ethnographic method of 'sensing data' by drawing on data as an ethnographic elicitation device, its implications and its uses. The study contributes to ongoing work that theorises big data as 'situated' (Rettberg, 2020) and 'mundane' (Pink et al., 2017). As Pink et al. (2017: 3) remind us, the mundane is not just unmarked 'ordinary' or everyday activity, but rather 'a generative site where people deal with contingency, improvise in the face of uncertainty'. This idea of mundane practices as a site where we can follow data being negotiated builds on an idea of data as being *situated*, a form of data analysis that has been stimulated by Haraway's (1988) call for situated knowledges, entailing an understanding of data as always being particular to its setting, never given nor objective, regardless of whether we are dealing with data on an individual level or handling immense quantities of data. If we think of the individual smartphone, computer or tablet user, my field note above encompasses what Rettberg (2020), in her description of situated data analysis, terms 'the personal level of data', that is personal data visualised for the individual user, but that can be shared with friends or other users. Correspondingly, when people use their phones, they can follow, either by contacting their data provider or examining their phone statistics, how much data they have used and for what purposes. Rettberg (2020) suggests analysing situated data at levels beyond the personal: Aggregated data visualised for humans, data aggregated for an operational dataset and data aggregated for machines to process, such as in algorithmic adaptation. As Rettberg points out, the most important analytic move in situated data analysis is to establish links between the individual and the aggregated – to understand, for instance, how the individual user's experience is interconnected with algorithmic adaptation. However, I focus here on a larger set of data circulating in telecommunication systems, rather than what has become a common site for mundane/situated data literature: Data produced within a particular app's context, such as Rettberg and others' studies on Strava (Lupton et al., 2018; Pink et al., 2017; Rettberg, 2020). My focus on telecommunication systems means that 'aggregated' sites look vastly different, as I will discuss below.

The ambition here, to develop new ways of knowing data, could be understood as an effort to contribute to the field of data literacy – or perhaps even 'big data literacy', a research field that often draws on information studies or governance literature (Sander, 2020). However, following Lupton et al. (2018), I prefer to emphasise the importance of sensorial data knowledge rather than erudite literacy 'to highlight a dimension of digital data that has rarely been

considered: their embodied and sensory nature' (Lupton et al., 2018: 3). Building on the digital sensory ethnography developed by Pink (2014, 2015), Lupton et al. (2018), in the same study, introduce 'data sense', a concept used to emphasise the role of human senses and emotions as being integral to knowledge and meaning making. In the present study, interest in the embodied and emotional aspects of how we make sense of data is expanded beyond the particular context of an app, with distinct attention paid to the researcher's own role: Rather than as an analyst of sensorial practices in others' data lives, I focus on problems and possibilities related to how we, as scholars, enter the field with our *own* sense of data. Furthermore, in the telecommunication systems context, relevant data sites beyond the personal do not necessarily take the form, as Rettberg (2020) suggests, of data aggregated in a particular visualisation, although these visualisations do exist, such as telecommunication system surveillance and monitoring. Instead, in an extension of sensorial data knowledge towards infrastructure studies, I argue that the most significant aggregated contexts interconnected with personal uses of phone data relate to the infrastructures that our data travel through – infrastructures that exist as interconnected local, national and global information systems.

My approach here therefore links the above literature on big data as situated, sensorial and mundane, with a tradition in science and technology studies of researching information systems' infrastructure ethnographically (Star and Ruhleder, 1996; Star, 1999). In particular, I draw on the concept of 'infrastructural encounters' and the attention it draws to the mutual co-shaping of situated everyday practices and infrastructural systems (Abildgaard, 2024). Inspired by Tsing's (2011) work on 'friction' – as particular sites and moments in which the everyday is made faster or is slowed down, enabled or immobilised – examining digital infrastructures through the lens of 'infrastructural encounters' is an attempt to overcome the difficult analytical task of encompassing infrastructures as 'things, but also the relationship between things' (Larkin, 2013: 319). This attention paid to the entanglement of the system and the everyday bears similarities with the recognition in data studies that 'the boundaries between small and big data are porous' (Lupton, 2016: 63). The methodology introduced here in this way inscribes itself into a phenomenological, as well as auto-ethnographic, tradition in infrastructure studies inaugurated by Leigh Star's (1990) observations about her own onion allergy and its relationship to problems associated with standards, invisible work, identity and marginality. Thus, studying digital infrastructures through the lens of infrastructural encounters calls for a phenomenological approach to technologies such as phones, Internet routers, fibre optic cables and satellites, as they are encountered from a first-person perspective at particular sites, while entangled with infrastructural systems on numerous scales.

A phenomenological engagement with digital infrastructures in Greenland also inevitably elicits questions about coloniality and marginalisation – not as stable, already-known power structures, but as they emerge concretely through infrastructural landscapes. An autonomous nation within the Kingdom of Denmark, Greenland's colonial relationship with Denmark is present in how digital devices, regulations and personnel circulate between both nations, with the framework of Greenland's infrastructures stemming from Danish-led modernisation (Abildgaard, 2022, 2023). This relationship is also painfully present during geopolitical crisis events, such as repeated offers and efforts from US President Trump, during both of his terms, to 'buy' or 'take' Greenland from Denmark. Importantly, I consider coloniality within the same analytical framework of infrastructure studies that informs my study of data. The challenge is not simply to confirm that infrastructures reproduce colonial and marginalising relations, but also to trace how these relations become tangible in specific encounters. Cardwell et al.'s (2024) observations on infrastructures as historically constituted, layered and recursive provide an important backdrop here, reminding us that infrastructures are not just material and practical, but also shape epistemic conditions.

In the following sections, I first detail this method of sensing data, as well as how and why it evolved, by taking you through selected data field notes, social media posts and interviews from Qaanaaq. Second, I discuss how to explore data sensing outside the context of ethnographic fieldwork and implement the method in a teaching context. Third, in a more reflective section on the method itself, I discuss potential uses and implications of sensing data. These three facets – autoethnographic experience, pedagogical experimentation and methodological development – are not separate contributions, but rather different modes through which sensing data, as an ethnographic method, can be practised, taught and theorised.

Fieldwork: notes on data

Before discussing the fieldwork itself, it is worth going a step past my earlier description of the shared 175 Mbit/s geostationary satellite connection that constitutes Qaanaaq's digital access to the world (and vice versa) and explain Greenland's digital infrastructure in broad terms. Greenland's telecommunication infrastructure often is described as comprising three zones: a cable zone; a radio chain zone; and a satellite zone (Pedersen, 2016). Two undersea cables, one running west to Canada and one east to Iceland, provide the fastest and only physical connection between Greenland's telecommunication infrastructure and the rest of the world. Both fibre-optic cables' lines land in Nuuk, Greenland's capital. Another sea cable extends this connection, travelling from Nuuk up along Greenland's west coast. This cable signal continues in a radio chain running farther north and south – a remnant of Greenland's pre-digital telecommunication infrastructure designed during

the Danish-led modernisation of Greenland mid-19th century. Outside of this infrastructure, towns and settlements depend on the slowest and most expensive digital link: a geostationary satellite connection. In this way, the farther you travel away from Greenland's cable zone, the slower your connection rates will be. You also will have fewer subscriptions to choose from, and they will be more expensive. In this national infrastructure, the most marked differences appear between the 'cable towns' (connected to the Internet through fibre-optic cable) on Greenland's west coast and the 'satellite towns' of the less-populated east and northwest (Abildgaard, 2023; Abildgaard et al., 2022).

The above story about Greenland's infrastructural systems was well-known to me, and I arrived in Qaanaaq prepared for a period of limited access to my email, my university's online learning system, social media, Zoom calls and all the other mundane practices of my life that require data. However, I had a hard time imagining precisely what 'limited' would mean. In preparation, I downloaded academic and fiction books on a Kindle and TV shows from different streaming services on a smartphone, and I bought a compact Switch gaming device for which I had bought the strategy game *Civilization*. An avid gamer, I had played and enjoyed *Civilization* when I was a teen and thought that my evenings in Qaanaaq would be a good time to revive that interest.

Travelling to Qaanaaq in northern Greenland from Denmark involves a journey on increasingly smaller aircrafts, first 3400 km northwest from Copenhagen to Kangerlussuaq in West Greenland, then 250 km from Kangerlussuaq north to Ilulissat. Ilulissat is home to the municipal head offices of Avaanaata, the municipality that covers Greenland's northernmost areas, including Qaanaaq and its surrounding smaller settlements. Still, Ilulissat is more than a thousand kilometres south of Qaanaaq. The trip continues 430 km north from Ilulissat to Upernavik for a brief stop before travelling 630 km farther north, finally arriving in Qaanaaq.

With this journey, I knew that the only option I would have to acquire a Greenlandic SIM card and phone plan would be Kangerlussuaq's modest Tusass store, which I had time to visit during a 24-hour layover. As the largest telecommunication company in Greenland, Tusass is government-owned and holds a monopoly on ownership and development of Greenland's telecommunication infrastructure, but has no local presence in Qaanaaq. In my field notes from that day, I briefly mentioned the visit and its outcome:

> **Day 1, 9/8**: *Arrived in Kangerlussuaq. Forgot my Kindle on the plane to Kangerlussuaq – no literature for the next few weeks. Wi-Fi at the hostel works well. I suppose the Tusass shop in Kangerlussuaq is my last opportunity for in-person advice on what to do for data while in Qaanaaq. Told the employee that I would not be using the Internet a lot; my needs were to respond to email and browse a bit. The advice was that I didn't need a subscription. Bought a Tusass SIM card and 200 DKK prepaid.*

As described in my notes, I explained to the Tusass employee that I would be spending a month in the northernmost areas and wanted to purchase a SIM card and some kind of subscription. When the employee asked what I needed, I did not have a good sense of what to answer, but mentioned that I would need to check email occasionally and browse the Internet briefly each day. What subscription did that warrant? The phone store employee thought that sounded like very minimal use, and recommended that I forgo a subscription entirely, as they could be quite expensive in the satellite areas, and just pay for the minutes and data I used with prepaid top-offs as needed. I therefore left the store with a Tusass SIM card with 200 DKK prepaid on it to use on data, phone calls and text messages.

Upon my departure for the plane northward the next day, an overview of my preparations indicated that aside from securing a SIM card with prepaid minutes, things were going less than ideally: I had forgotten the Kindle that contained reading material on the plane, and as I realised 2 hours before my plane for Ilulissat departed, I had *bought*, but not *downloaded*, *Civilization* – an immaterial difference in my everyday life that would prove to be quite an obstacle in the field. Having already checked out, I went back and sat for half an hour outside my Kangerlussuaq hostel room, using the last Wi-Fi available to me during the fieldwork before I needed to depart for the plane. The game managed to download 20%.

I arrived in Qaanaaq in the afternoon that same day. It was a mild August day, almost 10 °C, without much wind, and the sun appeared to be setting endlessly in the distance. Due to its northern position, Qaanaaq experiences a 24-hour 'midnight sun' during the summer months. I walked the 10 minutes downhill to the beach that brings the small town to a halt, as Qaanaaq lies between a sloping hillside and the water, and took a photo of the remarkable landscape. Standing there, I thought, 'Why don't I post this online – just this once – to convey that I have arrived safely and to let friends and family know that I will be out of reach for a while?' I posted the photo to Facebook and Instagram, and immediately received a notification from Tusass that 30 DKK of the 200 DKK was left in my account (Figure 1).

My first experiment with online life in Qaanaaq indicated that I had grossly underestimated how quickly my online activities during fieldwork would consume prepaid minutes. The following day, I walked to Qaanaaq's only store, Pilersuisoq, the local branch of Greenland's largest supermarket chain, which offers a reduced, and subsidised, selection of groceries and necessities in small or remote settlements. At the attached kiosk and post office, I asked if I

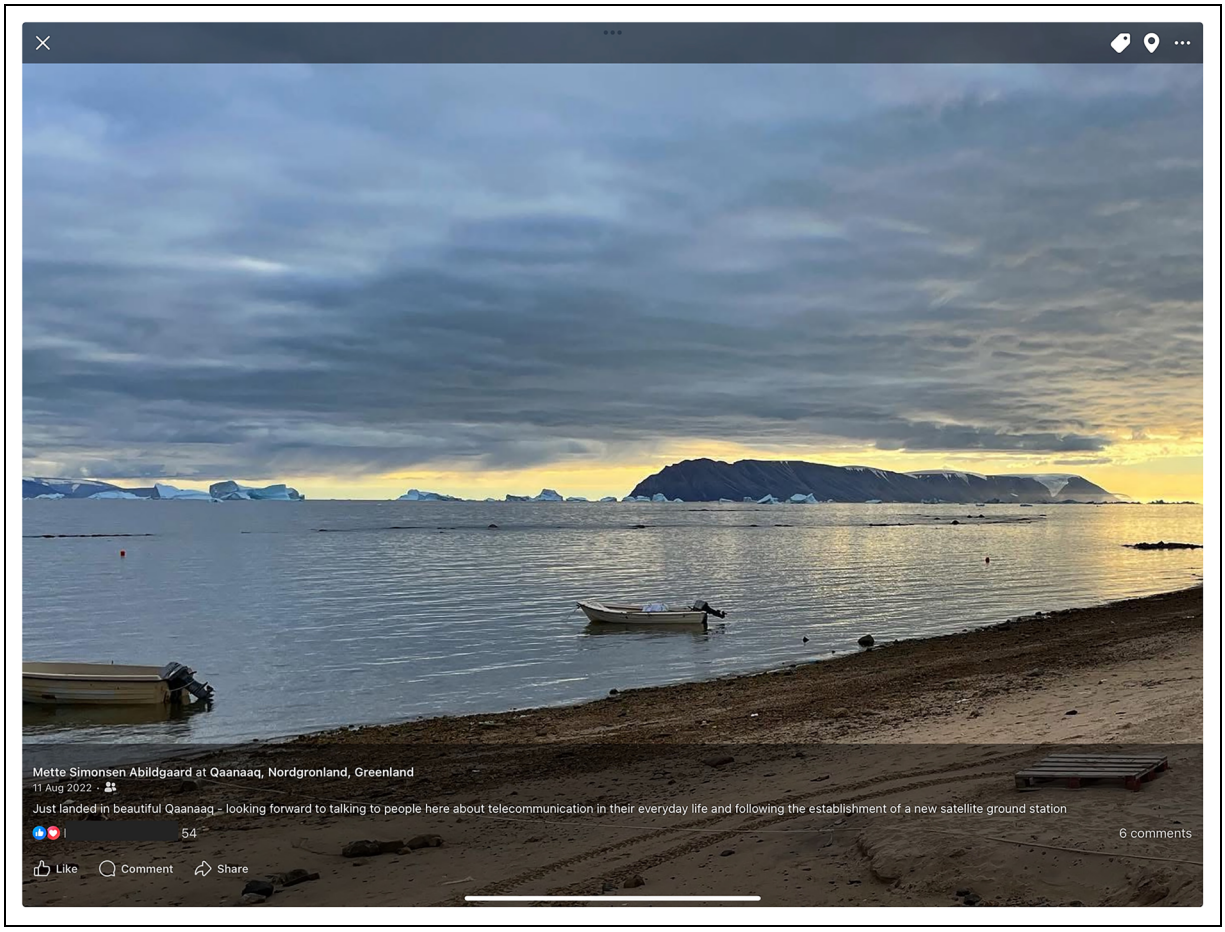


Figure 1. Author's post on Facebook from Qaanaaq (names of likers removed).

could purchase a Tusass subscription, but was directed either to use the Tusass website or phone customer service, as these were the only ways to sign up for a subscription. Therefore, I purchased an additional 100 DKK in prepaid minutes at the kiosk to call Tusass' customer service for advice. There, I was told that in Qaanaaq, you can choose between three subscriptions: 5 GB; 20 GB; or 50 GB of data. After the online post had used up almost all my prepaid data, I thought it wise not to choose the smallest package, so I ended up purchasing the 'most popular' middle option – a 20-GB subscription. For context, a high-speed Danish flat-rate subscription (with unlimited talk and data) at that time in 2022 could be purchased for under half the price of the cheapest subscription offered in Qaanaaq (Figure 2).

After purchasing the subscription, I experienced one of the recurring issues with Qaanaaq's satellite Internet connection, which, due to signal fading and solar winds, can leave the town without Internet for days. I spent a day frantically trying to coordinate the upcoming semester's teaching with colleagues in Denmark via text messaging. However, as I learned after trying to commiserate about the pervasive Internet issues with my host at the hotel, I was wrong to assume I was experiencing a typical locally recurring Internet breakdown: The Internet was working fine for everybody else. After changing my subscription, I just needed to utilise the universal solution of turning my phone on and off again to access the Internet. This finally left me, on Day 5 of my fieldwork, with telecommunications up and running. I even started transferring the remaining 80% of *Civilization*, letting the Switch download overnight.

I spent the next few days getting acquainted with Qaanaaq, talking with residents about their online practices. I downloaded a few weather (DMI), plane tracking (flightradar24) and wind (Windy) apps used in Qaanaaq to track weather and visiting vessels, and went online to RAL.dk to follow the cargo ship that will deliver Qaanaaq's second annual food supply in early September before the waters freeze. On Day 9 of fieldwork, I realised, as indicated in this article's introductory quote, that I already had spent 14 GB in 7 days, with eight more planned in Qaanaaq. As part of this over-spending, I used more than half a gigabyte of data on a function called 'systems services', which I had not noticed before and could not figure out what represented (Figure 3).

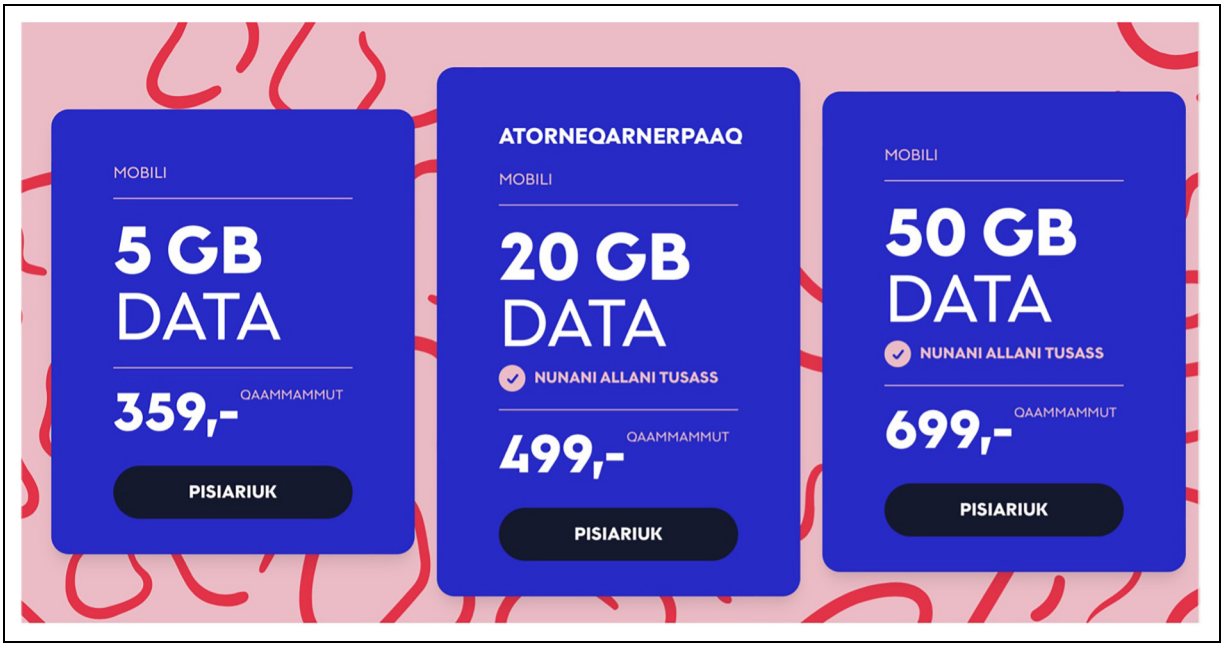


Figure 2. Screenshot of Tusass website displaying available Qaanaaq phone subscriptions and prices, by the author, 8/4/24.

This sudden realisation led to a crisis:

Day 9, 17/8: *14 GB of 20 used*

It's embarrassing that I have already almost spent my 20 GB, didn't think I used that much data on my phone. The kids use my hot spot all the time when we are on holiday without Wi-Fi, but I have to admit that I never thought about how much I have in my subscription at home. OK, shit, just checked – I have no limits on data and talk, of course. Turned off automatic updates of software. If I spend 1 GB a day, I will not even have enough. Gotta turn off mobile data at night.

Almost midway through my fieldwork in Qaanaaq, I was struck by the degree to which I lacked a sense of data. I had a vague notion that I was not, in a Danish context, a heavy data user, which I took to mean someone who did not browse video-based platforms like YouTube for hours on my phone and who had not, in recent years, tried running out of data. However, recognising that I had not even been aware of how much data my home subscription included, I began asking myself exactly how much data I spent doing everyday tasks, such as browsing the news online for half an hour, posting a photo online or doing an hour's video conversation with friends or colleagues. In particular, I realised I did not have a sense of how much data I continually used updating the software on my phone and computer – processes that usually would take place automatically on Wi-Fi at home or at work. Having spent a few weeks in Qaanaaq, I started getting notifications that these updates were overdue.

Therefore, I established a fieldwork protocol to help me get a better sense of data – a crucial skill to analyse how data is situated, as well as understand and navigate the mundane practices of an environment with limited access to data. Starting on Day 9, I logged my data use each morning by screenshotting my phone's mobile data overview, reflecting on what I had spent data on the day before. I also tried to control and minimise my data use while learning about and emulating, as much as possible, local practices of managing data.

So, what did situated data practices in Qaanaaq look like? Below, I list the most important local practices I encountered during my fieldwork and interviews in Qaanaaq (Figure 4).

In my understanding of these mundane routines for managing data, I am reminded of the concept of 'accounting practices', stemming from domestication studies, to describe a set of practices that are 'imbued with notions of who lets who use what, of moral judgements of the other's activities, of the expression of needs and desires, of justifications and conflict, of separateness and mutuality' (Silverstone and Hirsch, 1993: 104). Of course, contemporary data use is much more individualised than 20th century shared use of TVs and other media in households – the original loci of domestication studies. However, as Pink et al. (2017) note, mundane data practices are generative sites that involve conflict, and accounting practices highlight that they also involve morality, negotiation and care work (Mol, 2003, 2008). While more individualised, using data on a personal laptop or smartphone still involves an array of justifications regarding who gets to do what.

As I realised, my protocol from Day 9 in part already included local mundane data practices, as an important rule in a satellite town where data is limited and expensive is to

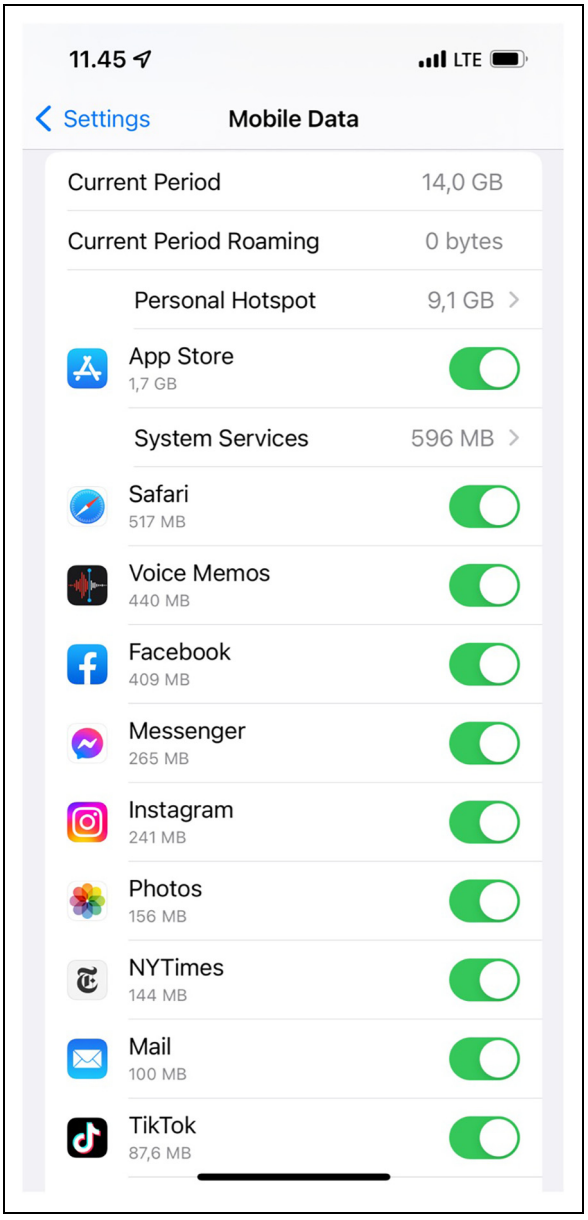


Figure 3. Screenshot of author's mobile data use, 17/8/22.

keep track of data. As Internet connection speed is impacted by the larger number of people online during daytime hours, another mundane practice I had taken up was to delegate heavy tasks for nighttime hours. I had also become more strategic about what I used data on, as I turned off data for apps I deemed non-essential. At the end of my 3-week stay, I settled on a practice in which I turned off mobile data entirely during nighttime hours to avoid spending data unnecessarily.

For myself, as well as the interlocutors whom I interviewed and spent time with in Qaanaaq, I also recognise the element of vanity and joy in seeing my own data spending and beating my record (of how little data I could use daily), a phenomenon that Lupton et al. (2018) found in a study of how cyclists sense their data. In my fieldwork notes from Day 14 (below), as well as an Instagram post from one of my interlocutors) (Figure 5), we can see different expressions of satisfaction at one's own data use score when using a sufficiently frugal data practice:

> **Day 14, 22/8:** 17.6 of 20 GB
>
> *Remembered to turn off mobile data during the night. Woke up early because of noise. Spent about 0.1 GB primarily on reading the news for about 30–60 mins. But it seems to work that I don't post images and limit my use to what is most necessary. I wonder whether people here also have the sense that it is the end of the month regarding data.*

Other practices were more difficult for me to engage with: in my fieldwork preparations, I had not anticipated using hard drives to store films and TV shows, so with my only downloaded shows on a private smartphone, I could not participate in their circulation. I also did not have visitors in the small room I stayed in at the town's hotel, nor did I have a sufficiently close relationship with any of the children in town that it felt fitting to 'gift' data by sharing some of mine.

Finally, I encountered some accounting practices because I overstepped them, committing social faux pas. During the interviews, my inquiries about mundane data practices were often met with hesitation, as I was asking someone living in a satellite town how to do fairly obvious things like how to pursue an interest in film (sharing films with friends on an external hard drive) or how to keep in touch with family living far away (Facebook, Snapchat or phone calls). However, some questions were met with a certain degree of suspicion, as when I asked a municipal employee whether their Internet access at work could help them save on private data:

> *I: If you can be on Wi-Fi here, while you are at work, then I guess you can save a bit of data*
>
> E: ….
>
> *I: Or perhaps that is not so?*
>
> E: I have a lot of respect for my place of employment.
>
> *I: Of course. If you are not allowed, that is how that is.*
>
> E: No. No, that is far too… from my phone, I use my own data when I am at work […]
>
> *I: I am just trying to understand because, I mean, at my work, I use Wi-Fi for all sorts of things, and if I am downloading an app or something, some music or something, then I don't think about it. I just use my work Wi-Fi … when I am in Copenhagen.*

Mundane data practices in Qaanaaq

Keep score: Keep track of your data use as the month progresses, or after you have used a significant amount of data.
Be strategic about WHAT: Prioritize and choose what you need to use data for, use 'lite' apps and turn off mobile data after you are done.
Be strategic about WHEN: The connection is often limited and slow during the day, so do heavy tasks during the night or be prepared to wait.
Reuse data: Keep your data use low by circulating films and television series on external hard drives among friends and family.
Do gift data: Data is valuable and limited, and can be gifted to visitors, especially children, who will appreciate visiting you
Don't steal data: Data is valuable and limited, using data privately from sources such as work is a form of stealing

Figure 4. Overview of mundane data practices in Qaanaaq.

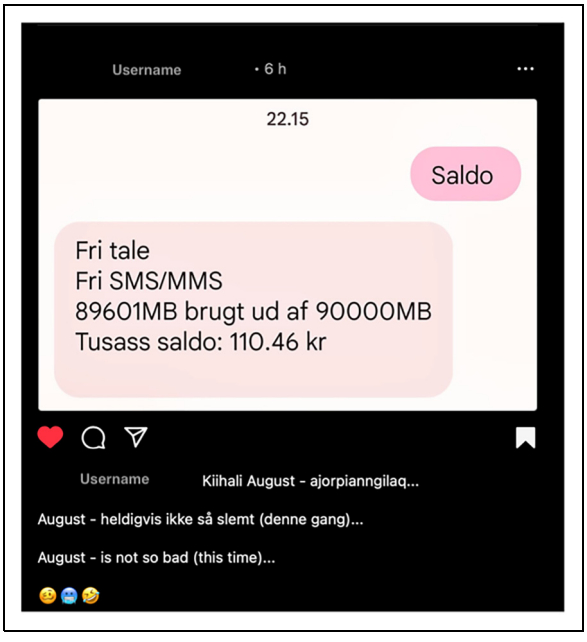


Figure 5. Anonymised Instagram screenshot of Qaanaaq citizen sharing the results of checking their remaining data for the month of August.

> *But I guess, of course, you have to think about that here, so that you don't delay or take up the Wi-Fi, I mean, in case you wanted to download an entire TV show at work.*
>
> E: No, that is not – you can't, we don't have the opportunity to do that. Our own private thing, that is our responsibility, not our employer's responsibility.
>
> (Interview, 2022. 'I' denotes interviewer, 'E' employee)

I had understood data's value and scarcity, and how it should be monitored and used carefully, but I had not followed the moral consequences of this: You can be generous and gift it, but you also can be immoral and steal it.

In this fieldwork account, I have presented, in a mostly descriptive manner, what it means to sense data. To take a preliminary step into reflection: With sensing data, I mean to denote a set of explorative attempts to understand data from a firsthand phenomenological perspective through engagement with 'data positionality', that is the researcher's embodied and emplaced (Pink, 2011) sense of data's character and shape. In the example above, a heightened sense of data was developed during fieldwork through meticulous accounting of my own data practices, as well as engagement in local practices in an unfamiliar infrastructural setting. Below, I continue in the same descriptive vein, but shift the focus towards a second experiment with sensing data, now in a small-scale teaching context.

The exercise: a metered mindset

The fieldwork described above comprises one way of ethnographically heightening and exploring one's sense of data, but fieldwork is also famously a prohibitively expensive and time-consuming method. In an effort to engage in more accessible ways of sensing data, in my own work as a university educator in Techno-Anthropology at Aalborg University in Copenhagen, I extended this form of engagement with data in the classroom through 'the metered mindset exercise'. The assignment is named after the 'metered mindset' concept, which Donner, drawing on fieldwork in South Africa and India, describes as a kind of attentiveness

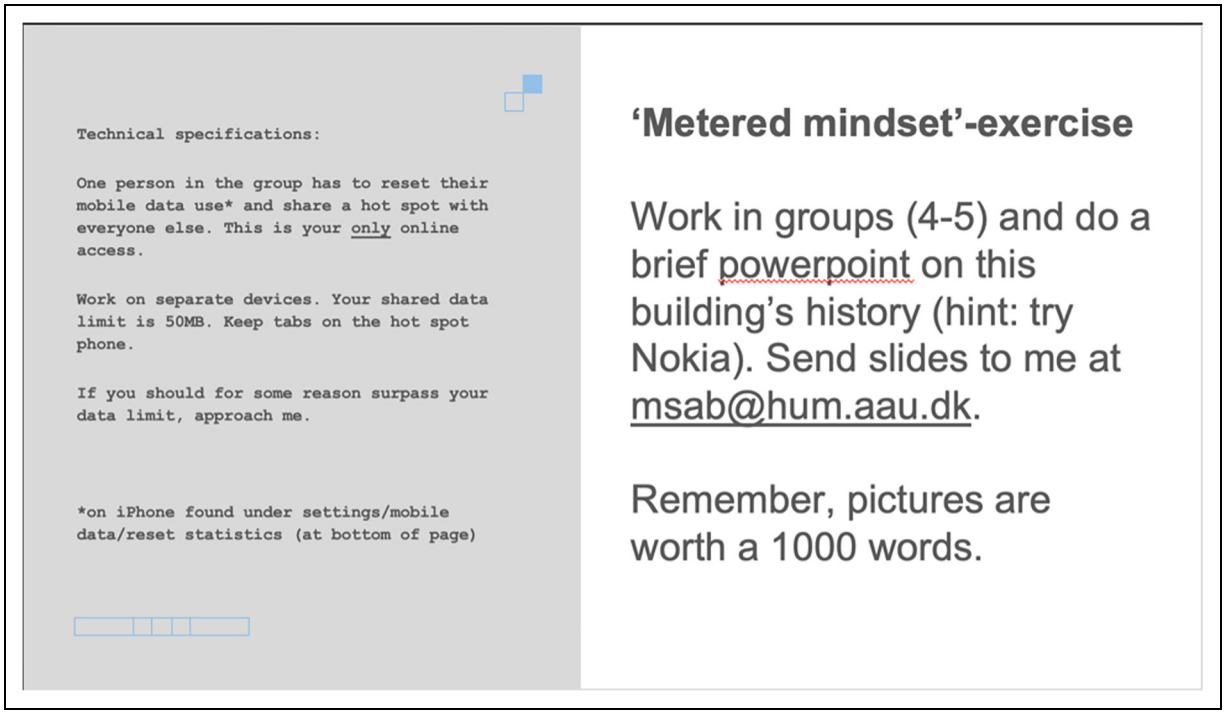


Figure 6. A slide from a presentation of the exercise.

that comes with realising the incremental costs of using mobile devices. Consequently, users monitor their use closely and 'do not browse and surf, they dip and sip, carefully conserving airtime and balances on their data bundles' (Donner, 2015: 135) (Figure 6).

In the 'metered mindset exercise', participants work in groups to research an unfamiliar topic and create an image-rich PowerPoint presentation. The key constraint is that they must complete the task using only one Internet connection – a mobile hot spot – with a strictly limited data allowance (set at 50 MB in my most recent iterations).[1] The hot spot phone has its data use reset beforehand for easy tracking.

Of course, the exercise's goal is not the creation of slides, but rather the process wherein participants bodily and emotionally sense their own data use and engage in metered mindset behaviour. Participants must negotiate strategies with teammates who may have different assumptions about how to conserve data, while still completing the task. Unlike my slowly emerging sense of data during fieldwork in Qaanaaq, a group-based approach quickly makes mundane data practices explicit. Different roles and tasks inevitably will emerge, such as tech experts (people who offer their sage advice on how devices work and which actions are data-free or costly), frugal consumers (people who draw on their personal experiences with low-data subscriptions), hotheads (people who forge ahead without much concern about data consumption) and worriers (people who urge everyone to go slowly, preserve data and avoid practices perceived as risky).

In many of the groups who have undertaken the exercise, the worrier role falls to the hot spot phone's owner, as they feel responsible for data consumption and will intervene and warn the group as the bytes tick by. In the below excerpt from the initial minutes of an exercise in one group, Signe is working with Anders (who steps into the role as tech expert) on her Mac computer to start the presentation, while Felix's phone functions as a hot spot:

***Anders**: You have to press Control-Alt-Escape, and then you can see all the background programs that are running.*

***Signe**: What should I push? You just push.*

***Anders**: No, it's Control-Alt-Delete. Isn't that it? No. It's Windows. No, it works here too.*

***Signe**: Fuck, I have a lot of shit!*

***Anders**: Then you can see all the programs that are running.*

***Felix**: We've already used 1.7 megabytes.*

***Anders**: OK, so we have 49 left. The considerations we're making now are that we start by interrupting all the background things that are running. To get the most out of our data.*

[one minute later]

***Signe**: Well, no, that's why there's a sauna here. It's because Nokia has founded…*[2]

***Felix**: We've already used 2.7 megabytes.*

***Anders**: Have we even started anything yet?*

***Felix**: We didn't even make a PowerPoint. If we have to send this to her too, we won't be able to do anything.*

A few aspects are apparent from this interaction. First, it is clear that when transitioning from Wi-Fi to a limited-data framework, the first minutes are crucial. If you do not keep track of your data, you will quickly run out of bytes just from all the 'background things' that connected devices continually run. The more devices connected to the hot spot, the quicker this process will happen – to the point where some groups ran out of data before even starting. Second, expertise is easy to undermine when it comes to mundane tasks involving data, as few people have detailed knowledge of what they involve. Instead, participants rely on what we might call 'data folk theories', such as how to conduct Google searches without encountering data-consuming images, whether sending an email is less data-costly than receiving it, or, as Rene speculated, what functions like 'power save' actually do: '*I'm pretty sure it managed to do almost all of that before you put it on power save. I'm pretty sure power save drops a lot of it. Right? It might not limit your data as forcefully as you'd think. But just a little bit*'.

Returning to my earlier point, the aim of sensing data is not to analyse a particular infrastructural level (e.g. the individual/consumer), but rather to encounter where everyday practices and infrastructural systems are interconnected. In this exercise, close engagement with data means that the infrastructural systems that phone data travel in become foregrounded, as we saw, for example, when participants noticed all the processes continually running on their phones. Below, a group comprising My, Rene and Louisa discussed how they ended up using 753 kilobytes on Louisa's iPhone within the first minute of the exercise. They realised that some of it was used on the apps Maps and Stocks, even though Maps was not open, and Louisa never uses Stocks. Two minutes later, the group assessed their data consumption again:

***Rene**: Are we even keeping an eye on it, or is it just using data on everything?*

***Louisa**: I don't know, I don't know, I can't really stop it, can I? We're at 15.*

***My**: OK, that's not so bad. It's only three more.*

***Rene**: But what the hell has it used it on since? These 3 megabytes.*

***Louisa**: Should we just turn all that off?*

[Louisa closes active applications on her phone]

***Rene**: Yes, maybe. Well, OK, you might have to have settings on.*

***Louisa**: Well, of course, because there's Internet sharing. What is it using now? OK, like that.*

***Rene**: What the hell has it been using since the last time?*

[they browse the list of apps that use mobile data on Louisa's phone]

***Louisa**: Well, since it keeps talking about it, I'm very interested in what I have in here myself. (giggles) That's not so good.*

***My**: Snapchat!*

***Louisa**: But I can – if I turn it off, I can just turn it back on.*

***Rene**: Yes, yes.*

***Louisa**: Should I turn it off now, all of this? Would you do that?*

***Rene**: Yes, maybe you should.*

***My**: (laughs) Are you on the 7/11 app a lot right now?*

***Louisa**: No, never (sighs).*

Here, inexplicable, apparently unstoppable data use leads the group to probe into what Louisa's phone is using their data on. The process ends with Louisa's exacerbated response to how apps such as 7/11, which she never actively uses anyway, nonetheless are spending their 50 megabytes. The exchange highlights that the exercise is not about identifying technical knowledge gaps, but rather having affective and practical experiences with data, such as exasperation at attempting to control a data infrastructure that operates so pervasively in the background, or the anxiety of limiting data from crucial lines of communication, such as Snapchat.

These vignettes from the metred mindset exercise demonstrate that engaging students in a hands-on exercise with metered data use can foster an embodied understanding of digital infrastructure as something more than an abstract system. Instead, it becomes something they physically sense and strategically navigate. As the participants struggled to conserve data, they became acutely aware of background data consumption and monitoring that usually are obscured. The exercise was followed up with a reflection session during which each group described its strategy, what they spent their data on and what gave them the most trouble. An interesting finding from my most recent iteration of the exercise was that what

participants found most troubling – and freeing – was having to do without many of the quickly found online facts (e.g. year of construction) that they viewed as essential for answering prompts such as those raised during the exercise, depending instead on their immediate surroundings, such as books and directly observable information (e.g. information plaques), and pooling existing information and hearsay from group members (*there are four saunas!*). The exercise ends with everyone present sharing an experience with data that they had during the exercise. In doing so, the 'metred mindset exercise' serves as a pedagogical tool that could be used in interview or focus groups settings as well, making infrastructural frictions and dependencies explicit and encouraging participants to reflect on their own digital habits and assumptions.

The method: sensing data

To repeat the issue, kilobytes, megabytes, gigabytes and petabytes are universally standardised matters, but the systems in which they travel are not. In the fieldwork and group exercise described above, we see how an ethnographic method of sensing data as mundane and situated takes form slowly, in improvised fits and starts. At the core of these methods is an effort to understand the relationship between individuals, their everyday and the vast quantities of data circulating within our telecommunication and information systems. Rather than approaching data as integral to a series of infrastructural levels to be analysed separately, this is about trying to capture encounters in which everyday practices and infrastructural systems are interconnected (Abildgaard, 2024).

The satellite infrastructures that characterise everyday life in Qaanaaq are my key example of this: Data becomes limited and, therefore, valuable, and particular practices of sharing – and protecting – data emerge. These practices reflect not just that subscriptions in Qaanaaq are relatively expensive, in a Greenlandic as well as global context, but also that the town's limited 175 Mbit/s satellite connection is shared with nearby settlements and other areas. This creates an infrastructural bottleneck during daytime hours that necessitates particular temporal data practices.

The teaching exercise above mirrors the ethnographic fieldwork. Both rely on direct, situated engagement with digital constraints, whether through immersive fieldwork or classroom exercises, but they operate at different scales and – potentially – for different audiences. I included both here to illustrate that I think the methodology of sensing data can speak to two kinds of readers: those engaged in ethnographic studies of global digital infrastructures, who are interested in how connectivity constraints shape everyday practices at specific sites; and those exploring ways to engage bodily and situationally with data, whether in pedagogical settings or in reflective research on digital habits. The former (researchers concerned with digital systems' materialities and discrepancies) may note the specifics of Qaanaaq's infrastructural landscape and data practices, tracing how infrastructures operate across geographic and political contexts. The latter might note how the exercise and the fieldwork illustrate universal struggles of adapting to a new infrastructural setting (e.g. my presumption that the satellite connection was the cause of my unconnected phone) and might include researchers, educators, designers or practitioners looking for methodological tools that allow for direct, experiential engagement with digital devices beyond conventional fieldwork.

In this way, the methodology of sensing data is clearly in conversation with several fields. Reflecting on the list of mundane practices in Qaanaaq that came out of the data sensing protocol, it forms a contribution to scholarship in human–computer interaction (HCI) on user experiences of bandwidth constraint (Donner and Walton, 2013; Sambasivan and Smyth, 2010). For example, in a study on technology use in Nairobi, Kenya, Wyche et al. (2010) found what they called 'deliberate interactions' – a planned and purposeful interaction style that involves offline preparation of online activities. Unsurprisingly, deliberate interactions also would describe the most successful student strategies for participating in the 'metered mindset' exercise. Looking across Nairobi, Copenhagen and Qaanaaq, recurring data practices point to what we might view as typical styles of interacting with data in areas of limited connectivity, such as practices of downloading larger files to a physical hard drive or CD, conducting offline research and thinking carefully about your actions before moving online to make the most of them.

Another pervasive link to studies on bandwidth constraints relates to frustration among interlocutors that collaborators elsewhere (e.g. in the United States) do not understand and account for local bandwidth limitations. These frustrations mirror the teaching exercise participants complaining about video- and image-heavy web pages consuming their data while they look for text, but most clearly the frustrations of my interlocutors in Qaanaaq when dealing with digital tools and systems 'from elsewhere'. In Qaanaaq's mundane data practices, we see an issue of tools unsuited for local bandwidth conditions, reflected in the use of alternative 'lite' versions of apps that are globally popular, such as Instagram or Facebook. Unfortunately, as I have documented more extensively in other studies, when Qaanaaq's residents deal with difficult online-only software introduced by the municipality farther south, or digital devices imported from Denmark, alternatives are rarely available (Abildgaard, 2024).

The methodology also resonates with studies on digital practices in constrained environments beyond bandwidth limitations. Burrell's (2012) ethnography of Internet cafés in Ghana illustrates how young people navigate connectivity constraints through communal approaches to data use. Similarly, Tacchi et al. (2003) argued for the importance of ethnographic action research in studying digital media

use among marginalised communities, demonstrating how limited access shapes everyday digital habits. These studies reinforce the argument that data practices are not random or indiscriminate interactions, but emerge in dialogue with infrastructural conditions.

Approaches in anthropological studies on media uses and practices (e.g. Moores, 1993; Morley, 1986; Silverstone and Hirsch, 1993) also offer interesting links to the methodology of sensing data. Studies among user groups with low socioeconomic status are a reminder that conditions that impact data practices go beyond telecommunication infrastructures and areas with low connectivity. One study found that Italian teenagers developed everyday strategies, such as hopping from Wi-Fi to Wi-Fi – a strategy that is not feasible in Qaanaaq. However, other strategies, such as checking mobile data settings or texting one's provider to monitor use, resonate across contexts (Drusian et al., 2022). In this way, the method of sensing data connects to broader discussions on digital resourcefulness, tinkering and adaptation.

However, unlike the predominant literature on media use and user bandwidth constraints, the method of sensing data was developed first and foremost as a way to understand one's own data positionality, an element that I argue is key to ethnographic data work in a double sense: To produce ethnographic data in ways that capture how data is situated. First, it highlights how we can produce ethnographic knowledge about data in ways that account for its situatedness. Second, in the absence of a 'standard' or 'correct' information systems infrastructure, an understanding of positionality is a central component of any ethnographic data study and is critical in avoiding unexamined assumptions about what constitutes appropriate data practices in a given field.

We saw a striking example of such imported assumptions when I inadvertently suggested that a public employee in Qaanaaq stole their employers' data. This interpretation, rooted in my own experiences with data consumption as a resident of one of the world's most digitised countries, introduced distrust into the conversation, rather than foster a discussion of data practices at work: Why was I making such suggestions? Would I rat out my interlocutor to their boss? While we could frame this as a function of my Danish background in a postcolonial setting, I would argue that the more immediate positional factor was a contrast between my urban life in Copenhagen – where Wi-Fi is ubiquitous and inexpensive – and the practicalities of satellite town infrastructures, where data access is more constrained.

This distinction between colonial and urban positionality matters. While past Danish infrastructural governance and imported digital technologies shape Qaanaaq's bandwidth limitations, Greenland's digital infrastructure has developed during 46 years of home- and self-rule. As a result, Qaanaaq's most pressing asymmetries emerge in relation to closer systems – such as the municipal offices in Ilulissat or the headquarters of national telecom firm Tusass in Nuuk – rather than relics of colonial governance (Abildgaard, 2023; Abildgaard, 2024). In this way, recognising data positionality helps foreground digital practices' generative and situated character, rather than reduce them to particular narratives.

The method of logging data use in detail described here also shares a link with 'data diaries' (Tkacz et al., 2021). However, while data diaries aim to follow data as an isolated phenomenon, refraining from personal perspective and narration, to *sense* data, phenomenological reflection is key, keeping this methodology closer to the ethnographic diary genre in a stricter sense.

In particular, the methodology owes a debt to an ethnographic body of work that began with Malinowski's (1989) posthumously published diaries, which stand out for their intimate portrayal of their author's personal doubts, frustrations and hopes, which contrast with the degree of certainty and authority projected in classic ethnographic writing, including that of Malinowski. While Malinowski never intended for these diary excerpts to be included in his published works, later developments in anthropology – such as the crisis of representation, narrative turn, feminist and postcolonial scholarship, and autoethnography – have unsettled the empiricist ideals that used to structure ethnographic authority. As Scott et al. (2012) note, these shifts have given rise to 'a new contingent repertoire of tacit knowledge, social pressures and institutional politics' that contrasts with the 'formal, sanitised accounts of data collection' that characterised these earlier traditions.

Crucially, as Wästerfors (2023) suggests, this turn towards the embarrassing and emotional in ethnography rarely positions the researcher in a 'strong, powerful or paternalistic position in relation to the researched'. Instead, as researchers, we often are navigating emotional and dramaturgical dilemmas in our attempt to perform the role of competent fieldworkers. To this end, sensing data entails engaging in ethnographic observations and interviews in a way that foregrounds the misunderstandings and disturbances we cause, and the risks entailed in inquiring about situated data practices. I also think of this attention to difficulty and embarrassment as a tool to counteract the tradition of 'formal, sanitised' accounts produced by researchers in positions of power, requiring instead that we provide a degree of vulnerability similar to what we ask of interlocutors (Punch, 2012: 91), an important guideline for researchers, such as myself, undertaking fieldwork in a colonial context. Graugaard (2021), reflecting on her autoethnographic work in Greenland, suggests that such autoethnographic practices, while not necessarily decolonial in themselves, can 'provide reflections on ways to destabilise (our) (neo)colonial research practices'. Rather than concealing the frictions and uncertainties of fieldwork, sensing data foregrounds them, making them central to understanding how data circulate, acquire value and are negotiated in emotional, situated and infrastructural contexts.

I have been talking about data here in a dual sense – as an ethnographic method as well as an object of inquiry – discussing the implications of drawing these two together. Looking across these two forms of engagement with data, a particular kind of attention or register emerges – an attention to limits and an acknowledgement of that which is not smooth nor easy. To sense data entails that we disturb the field and risk ourselves, asking stupid and insulting questions. This disturbing practice also accentuates the limits, risks and difficulties of another kind of data – that which circulates in digital systems.

While extant studies in this tradition of situated, mundane data studies have sought to understand, for example, how the individual user experience is interconnected with algorithmic adaptation (Rettberg, 2020), I seek to bring attention to the information infrastructures in which such data practices take place. How data are valuable in Qaanaaq is particular to *this* site, but it also allows us to perceive a global issue, as it draws attention to the previous decade's increasing availability of 'unlimited' flat-rate subscriptions.

The rise of 'unlimited' data plans,[3] marketed as subscriptions that include unlimited texting and calls, as well as unlimited Internet access (although typically limited to the subscriber's home country) can create the sense that information infrastructures are irrelevant to situated, mundane data practices. After all, who today thinks about how and when they use their data? However, access to affordable unlimited phone plans is distributed very unevenly worldwide. For example, a Statista survey revealed that high-income Western countries – such as Finland (61% of respondents), Switzerland (53%) and the United States (46%) – top the list of smartphone users with unlimited plans, whereas 22% of respondents from India were listed as having an unlimited plan (Zandt, 2021). In places with a lack of infrastructural redundancy and dispersed population patterns, such as Greenland, unlimited phone plans often do not exist.

Indeed, the presence of a 'mobile service' menu on the iPhone used during my fieldwork (see Figure 3), which promises the user that they can 'Find out how much data you're using [and] set data restrictions' (iOS Mobile Service menu, 2025), suggests that I, my interlocutors in Qaanaaq and the metered mindset-participants are not alone in grappling with the need to monitor and constrain data use. The fact that such settings are built into billions of devices across iOS, Windows and Android systems reflects that the smartphone industry not only acknowledges this need, but also actively sets an expectation that users should take responsibility for managing their data consumption. Such functionalities position the lay user as an independent, resourceful (micro)manager of digital resources – a role that aligns with broader narratives of personal responsibility in technology use (Mattson et al., 2017).

Notably, these data monitoring features' designs determine what is available to the user. While they provide an encouraging dashboard, they also make it difficult to fully track or regulate consumption. As in the case of my attempts at data management in Qaanaaq, opaque categories, such as 'system services', accounted for a high share of data use, without easy explanation or access to regulation. Data-monitoring in this way plays into mundane data practices by providing users with a sense of control, but also risks assigning users accountability for 'managing their data' despite severely limited agency and insight into its pathways.

So, the 'unlimited' plan suggests that data is a boundless, inconsequential resource. As we have seen here, this is not true in relation to users, or their subscription plans, nor is it true in relation to the infrastructures or planetary resources that these plans depend on (Çelik et al., 2024), and it risks increasingly limiting our sense of data. Contrary to the promise of unlimited data, we face a fragile, limited and uneven infrastructure. As scholars in 'broken world thinking' and infrastructural ruination have argued, erosion, breakdown and decay are a more fitting ontology for the world we inhabit than innovation, growth and progress (Anand et al., 2018; Howe et al., 2016; Jackson, 2014). Living within such decaying and limited digital systems necessitates paying attention to care work, such as the mundane data practices at play in Qaanaaq.

Conclusion

In this article, I introduce sensing data, a methodology that builds on phenomenological autoethnography to engage with data as embodied and situated phenomena. The approach extends existing scholarship on data as situated (Rettberg, 2020) and mundane (Pink et al., 2017), but shifts our focus from the context of a particular app to broader infrastructural engagement with kilobytes, megabytes, gigabytes and petabytes circulating through mobile devices, undersea cables and Internet routers.

During fieldwork in Qaanaaq, Greenland, sensing data emerged as a way of systematically reflecting on my own, as well as local data practices, and engaging with infrastructural constraints. It offers a means of encountering infrastructure (Abildgaard, 2024), tracing how everyday practices and information infrastructures mutually shape one another. A second experiment with sensing data unfolds in the metered mindset exercise: by imposing strict data constraints, the exercise tasks participants with negotiating a digital strategy in group work, foregrounding the often-invisible infrastructural processes that shape online access. Rather than aiming to foster data literacy in a conventional sense, the exercise highlights the affective and practical dimensions of navigating new infrastructural constraints, prompting participants to reflect on their own data habits and assumptions.

From these experiments, I propose that developing a sense of data is the key to recognising our data positionality – the often-unexamined assumptions about proper data

practices that stem from our embodied, situated experiences of digital infrastructures. Processes of sensing data not only foreground how researchers perceive and interact with data, but also amplify misunderstandings, disturbances and risks inherent in studying situated data practices. Such disruptions are particularly relevant when conducting research in a (post)colonial fieldwork context, in which traditional methodologies risk producing formal, sanitised accounts that obscure how data circulate, acquire value and are contested within embodied, situated infrastructural contexts.

Furthermore, while the method of sensing data goes part of the way towards underscoring how infrastructural constraints complicate the idea of seamless, individualised control of data consumption, future studies on how the smartphone industry conceptualises users as data managers can hopefully explore this more fully.

The method of sensing data might at first glance appear to align with approaches of studying what has been called 'the digital divide', that is the division between the haves and have-nots in digital infrastructures (van Dijk, 2006). However, I do not seek to highlight connectedness as an either/or condition, an ideal or even state of normalcy, but instead aim to offer a way of studying situated data in fundamentally limited/broken infrastructures. Drawing on the literature on infrastructural ruination and broken world thinking, I emphasise that the 'normal' condition is not smooth functionality, but rather a brokenness that requires constant repair and care work (e.g. in the form of mundane data practices), rather than innovation and growth (Anand et al., 2018; Howe et al., 2016; Jackson, 2014). To engage with and understand life in broken and limited digital infrastructures, we need a sense of data.

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Consent to participate

Participants in this study gave written consent to participate.

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Notes

1. While writing this paper, I repeated the exercise with three groups of three Techno-Anthropology students who gave their informed consent to participate in this study. All direct quotes in this section stem from this iteration of the metered mindset exercise, and the students' names have been anonymised.
2. Aalborg University in Copenhagen is housed in the former headquarters of Finnish mobile giant Nokia and therefore contains a surprising number of saunas.
3. Of course, the notion of the 'no limits' phone plan is itself a misrepresentation, as studies as well as a close reading of the terms of such subscriptions will demonstrate: After a certain threshold of data use, providers will throttle, i.e., intentionally reduce data transmission rates (Bauner and Espin, 2023).

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